Original article

Nuclear Medicine Communications

Optimized radioiodine therapy for Graves' disease: Two MIRD-based models for the computation of patient-specific therapeutic ^{131}I activity

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Aim **^{131}I therapy is increasingly used for Graves' hyperthyroidism. Debate remains about the best method for calculating the activity to administer, as well as about the potential benefit of such computed activity. Several arguments plead, nevertheless, in favour of a personalized computation, such as inter-individual variations of thyroid volume and biokinetics.**

Methods **A MIRD-based dosimetric approach, with an additional extension that takes into account the variation of thyroid mass during the treatment, has been developed. This approach includes the benefits of a personalized determination of biokinetics. Results were compared with those of six methods widely used in routine practice. Forty-one patients were enrolled (34 women, seven men; mean age ± SD: 48.11 ± 6.4 years). ^{131}I uptakes were measured at 4, 24 and 96 h (36.2 ± 14.6%, 42.8 ± 9.7% and 27.6 ± 6.8%, respectively), following administration of the tracer. The kinetics of iodine in the thyroid were evaluated using a two-compartment model (effective half-life of 5.1 ± 1.6 days). Computations of activities to deliver the doses prescribed by the physician were done with the eight formalisms.**

Results **There was no statistical difference between results of the two MIRD-based formalisms (227 ± 148 MBq and 213 ± 124 MBq), which were also not significantly different from those obtained with the majority of the other methods (from 128 ± 95 MBq to 275 ± 223 MBq). However, a large intra-individual difference up to a factor of 2 between two given methods was found.**

Conclusion **The formalism developed appears to be a good compromise between all the common formalisms already used in many institutions. Furthermore, it allows the exposures of target volumes and non-target volumes to be planned individually and practical individual radiation protection recommendations to be implemented.** ***Nucl Med Commun*** **27:559–566 © 2006 Lippincott Williams & Wilkins.**

Nuclear Medicine Communications 2006, **27**:559–566

Keywords: **^{131}I, Graves' disease, therapy, MIRD-based models**

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Received 24 January 2006 **Accepted** 30 March 2006

Introduction

Surgery and ^{131}I are the two classical therapeutic options for relapsed Graves' disease, and in many industrial countries, radioiodine has largely replaced surgery because of its safety, convenience and effectiveness. However, there is still controversy about the choice of the best therapeutic ^{131}I regimen, between an empirical, fixed activity and an activity calculated with a dosimetric method. Factors may influence the outcome of radioiodine treatment, such as the importance of thyroid goitre volume, severity of hyperthyroidism, combined antithyroid drug regimen, and individual target tissue sensitivity. Thus, it is not surprising that several studies have reported conflicting results about the ability to predict treatment failure or success with dosimetric methods [1–4]. Likewise, despite numerous methods, it remains impossible to accurately determine the optimal ^{131}I activity to cure hyperthyroidism while avoiding a permanent secondary hypothyroidism. However, hypothyroidism is not regarded as a serious complication, because of the effective use of thyroid hormone replacement, and even corresponds to the natural long-term evolution of Graves' disease, irrespective of the amount of radioactivity delivered [5,6].

Nevertheless, the administration of a fixed activity may lead to too much or too little radioactivity being delivered; that is, unnecessary exposure for the patient and for the environment or, conversely, insufficient ^{131}I to effect a cure. Furthermore, because younger patients are now offered radioiodine therapy earlier in the course of their disease, concerns about the long-term effects of

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radioiodine and thyroid hormone replacement therapy suggest the need of caution with large ablative doses of ^{131}I. Therefore, the optimal approach should be the administration of a computed therapeutic activity that takes into account individual parameters, such as gland size, iodine uptake and iodine turnover. A few studies emphasize that the absorbed dose to the thyroid is related to treatment outcome [7,8] and that only personalized dosimetry is able to take into account some particular cases where non-optimized dosimetry can be inefficient and/or hazardous [9]. However, the choice of the most appropriate dosimetric methodology to calculate therapeutic ^{131}I activity, according to a desired dose to the thyroid, is still highly controversial [7]. Thus, it could be useful to develop a standardized method where the latest advances in internal dosimetry and individual biokinetics are taken into account. An important point must be the reproducibility and the applicability in routine work in a wide range of nuclear medicine departments. In the present study, a formalism based on the MIRD methodology was developed as personalized dosimetry. One of the requirements was the facility to implement it in clinical routine. Results obtained with this new MIRD-based approach were also compared with those derived from other methods widely used in clinical practice.

Patients, materials and methods

Patients

This study involved two institutions: the nuclear medicine departments of Brest and Nantes University hospitals. A total of 41 consecutive patients (48.1 ± 16.4 years) suffering from Graves' disease were included (34 women and seven men).

Initial ($n = 34$) or relapse ($n = 7$) Graves' thyrotoxicosis were diagnosed on the basis of clinical findings and laboratory tests showing high value of free thyroxine and low levels of thyroid stimulating hormone for more than 3 months. Relapses occurred after at least 18 months of antithyroid drug treatment. All patients were instructed to stop the antithyroid medication 7 days before undergoing ^{131}I uptake measurements.

Uptake measurements methodology and experimental set-up

The patients received a capsule of ^{131}I (2 MBq) as an oral test activity. The ^{131}I capsule was measured at $t = 0$ and further thyroid uptake measurements were performed after 4, 24 and 96 h.

Measurements were performed with a $2'' \times 2''$ NaI(Tl) detector connected to a multichannel analyser. The energy window was set to 312–435 keV. Measurements were corrected from room and patients' background. They were performed at a constant distance (25 cm) between the detector and the thyroid (or the capsule for the first measurement). Three independent acquisitions (by displacing and replacing the probe) were done for each counting. The thyroid uptake was set to be the mean count of the three measurements. Additionally, all measurements were performed by the same technologist of each institution in order to minimize the daily probe mispositioning. The detector sensitivity was checked with a ^{133}Ba source each day prior to patient investigations and a correction factor was applied if a counting deviation $> 5\%$ was noted; that is, if the ratio between the first and the following ^{133}Ba measurements was > 1.05 or < 0.95, the uptake value was corrected by multiplying the measurement with this ratio.

Measurement of thyroid mass

The thyroid mass was evaluated by using two-dimensional ultrasonography methodology. The ellipsoid approximation based on height (h), width (w) and depth (d) of each thyroid lobe was employed. Assuming a density of 1 kg·cm^{-3}, the thyroid mass is given by the relation

$$m = \frac{\pi}{6} \times h \times w \times d.$$

Basic theory of the methods used for computing the activity to be delivered

Eight activity computations were chosen: three are based on the formula given by Marinelli *et al.* [10] (so-called Marinelli 1, Marinelli 2 and Marinelli 3), two were described by Leslie *et al.* [1] (so-called Leslie 1 and Leslie 2), one was based on the original work of Bockisch *et al.* [11] and one was developed in this study with an additional extension which takes into account the mass variation during the treatment (approach 1 and approach 2). The two last methods used modelling of the thyroid biokinetics and the MIRD formalism.

It should be emphasized that the Leslie 1 and 2 methods do not incorporate information regarding the dose delivered to the thyroid. These methods are only based on a fixed activity administration weighted by the thyroid mass and the 24 h thyroid uptake.

Marinelli computation

The Marinelli 1 method

The activity to be administered is given by the equation

$$A = \frac{m \times D}{73.8 \times E \times F(t=0) \times T_{\text{eff}}} \qquad (1)$$

in which A is the activity to be administered (in μCi), m is the thyroid mass (in g), D is the absorbed dose (in cGy) delivered to the thyroid, $F(t = 0)$ is the thyroid uptake (as a percentage) extrapolated to time $t = 0$, T_{eff} the effective half-life (in days), E is the mean electron energy ($E = 0.195$ MeV) and 1/73.8 is the factor which puts the equation into units of microcuries. The value of $F(t = 0)$ is deduced by linear regression over all thyroid uptake

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points corrected from radioactive decay. T_{eff} is computed from an exponential fit including the experimental thyroid uptake (non-corrected from radioactive decay) from 24 to 96 h.

Only the electron energy loss is taken into account in Equation 1. Marinelli assumed that the photon contribution was no more than 10% of the electron contribution. This assertion is added in our calculation ($D = D_{\text{electron}} + D_{\text{photon}} = D_{\text{electron}} + 0.1D_{\text{electron}}$).

The Marinelli 2 method

The only difference between the Marinelli 2 and the Marinelli 1 methods is the determination of the effective half-life.

$$A = \frac{23.4 \times m \times D}{F(t=0) \times T_{\text{eff}}} \tag{2}$$

where 23.4 is the factor that puts the equation into units of MBq, T_{eff} is equal to 5 days if $F(t=4) F(t=24)$ and 4 days if $F(t=4) \leq F(t=24)$, where $F(t=4)$ and $F(t=24)$ are the thyroid uptake measured at 4 h and 24 h, respectively, after oral administration.

The Marinelli 3 method

In this method, the activity is given by the equation

$$A = \frac{23.4 \times m \times D}{F(t=0) \times T_{\text{eff}}} \tag{3}$$

where 23.4 is a factor that puts the equation into units of MBq, $F(t=24)$ is the thyroid uptake at $t = 24$ h (corrected from the radioactive decay) and T_{eff} is equal to a fixed value (5 days).

The Leslie method

Two of the original propositions by Leslie *et al.* are retained [1]. This calculation is based on the administration of a fixed activity weighted by the thyroid uptake at 24 h ($F(t=24)$) and the thyroid mass. The first formula is defined as 'low-adjusted' (Leslie 1):

$$A = \frac{2.96 \times m}{F(t=24)}.$$

The second formula is defined as 'high-adjusted' (Leslie 2):

$$A = \frac{4.44 \times m}{F(t=24)}$$

in which 2.96 and 4.44 are two constants expressed in $\text{MBq} \cdot \text{g}^{-1}$ and $F(t=24)$ is the thyroid uptake at 24 h corrected from the radioactive decay.

The Bockisch method

This formalism is based on the thyroid uptake measurement at 96 h [11]. The activity to be administered is given by the relation

$$A = \frac{2.46 \times m \times D}{F(t=96)}$$

where 2.46 is a constant expressed in $\text{MBq} \cdot \text{g}^{-1} \cdot \text{Gy}^{-1}$ and defined over a population of 175 patients [11], and $F(t=96)$ is the thyroid uptake at 96 h without a correction for radioactive decay.

Two-compartment modelling associated with the MIRD formalism

The average dose delivered to a target due to activity embedded in a source region $\bar{D}_{\text{target} \leftarrow \text{source}}$ is given by the well-known relation

$$\bar{D}_{\text{target} \leftarrow \text{source}} = \tilde{A}_{\text{source}} \times S_{\text{target} \leftarrow \text{source}} \tag{4}$$

where $\tilde{A}_{\text{source}}$ is the cumulated activity in the source, and $S_{\text{target} \leftarrow \text{source}}$ is the S factor (i.e., the dose per unit of cumulated activity).

We assumed that the absorbed dose in the thyroid was only the consequence of the radioiodine uptake (there was no contribution from other possible non-target organs). Besides, we supposed a uniform distribution of ^{131}I in the thyroid. In this case, the source and the target are identical; therefore the S factor is noted as S.

Two S factors extracted from the work of Cristy and Eckerman [12] are considered in order to basically distinguish the computation between an adult male ($S_{\text{male}} = 1.57 \times 10^{-6} \text{Gy} \cdot \text{MBq}^{-1} \cdot \text{s}^{-1}$) and an adult female ($S_{\text{female}} = 2.59 \times 10^{-6} \text{Gy} \cdot \text{MBq}^{-1} \cdot \text{s}^{-1}$). Two thyroid geometries are considered: one for male ($m_{\text{ph}} = 20.7$ g) and the second for female ($m_{\text{ph}} = 12.4$ g). More realistic female anthropomorphic phantoms (as those developed by Stabin *et al.* [13]) could have been considered. However, the multiplication between the corresponding S factors (found in MIRDOSE3 software [14]: $S_{\text{female}}^{\text{MIRDOSE3}} = 1.90 \times 10^{-6} \text{Gy} \cdot \text{MBq}^{-1} \cdot \text{s}^{-1}$) and the new mass ($m_{\text{ph}} = 17$ g for the female phantom as described by Stabin) leads to almost similar results (0.6% difference).

The S factor should be modified to take into account the thyroid patient size. The correction factor for the non-penetrating component is the ratio of the phantom mass to the estimated actual mass. Thus, the new S factor, S^{patient}, is related to the S factor determined by Cristy and Eckerman [12]:

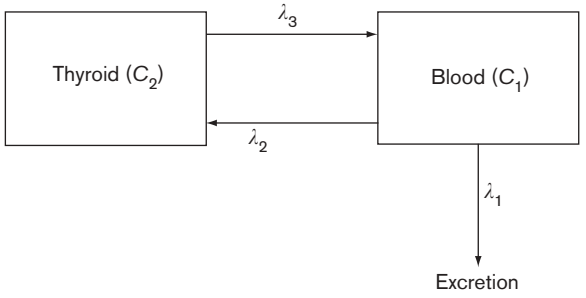
$$S^{\text{patient}} = S \frac{m_{\text{ph}}}{m} \tag{5}$$

where m, the thyroid mass, is evaluated by two-dimensional ultrasonography.

Both penetrating and non-penetrating components are included in the S values but the penetrating component is only about 3% of the total [15] thus justifying the use of the mass-corrected S value.

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Fig. 1



Simplified two-compartment modelling.

The biokinetics were evaluated by a standard two-compartment model (Fig. 1), extracted from the three-compartment model [16]. The compartment which describes the quantity of radioiodine taken from the body and eliminated by faeces was neglected as suggested by Di Martino *et al.* [17] and Matheoud *et al.* [18].

The resolution of such a two-compartment system leads to the following differential system:

$$\begin{cases} \dfrac{dC_1}{dt} = \lambda_1 C_1 - \lambda_2 C_1 + \lambda_3 C_2 \\ \dfrac{dC_2}{dt} = -\lambda_3 C_2 + \lambda_2 C_1 \end{cases}$$

where C_1 and C_2 are the concentration activities of radioiodine in blood and thyroid and λ_1, λ_2, and λ_3, are the kinetics parameters of the model.

Assuming $C_1(0) = A_0$ and $C_2(0) = 0$, the system was solved by the Laplace transform:

$$C_2(t) = \frac{\lambda_2 A_0}{\alpha_1 - \alpha_2}[\exp(\alpha_1 t) - \exp(\alpha_2 t)] \quad (6)$$

where

$$\alpha_1 = \frac{-(\lambda_1 + \lambda_2 + \lambda_3) + \sqrt{(\lambda_1 + \lambda_2 + \lambda_3)^2 - 4\lambda_1\lambda_3}}{2}$$

and

$$\alpha_2 = \frac{-(\lambda_1 + \lambda_2 + \lambda_3) - \sqrt{(\lambda_1 + \lambda_2 + \lambda_3)^2 - 4\lambda_1\lambda_3}}{2}.$$

The first formulation (approach 1)

The relation between $\tilde{A}$ and the initial activity A_0 is given by the equation

$$\tilde{A} = A_0\tau \quad (7)$$

where τ is the residence time. τ is expressed by the relation

$$\tau = \int_0^{+\infty} C_2(t)dt. \quad (8)$$

By incorporating Equation 6,

$$\tau = \frac{\lambda_2}{\lambda_1\lambda_3}. \quad (9)$$

Thus, the activity to administer A_0 is given by the equation

$$A_0 = \frac{\lambda_1\lambda_3\bar{D}}{\lambda_2 S^{\text{patient}}} \quad (10)$$

$$\Leftrightarrow A_0 = \frac{\lambda_1\lambda_3\bar{D}m}{\lambda_2 S m_{\text{ph}}} \quad (11)$$

where $\bar{D}$ is the prescribed dose to the thyroid. The thyroid mass is assumed to be constant during all the treatment. This relation is the statement for approach 1.

The second formulation (approach 2)

Many authors have reported a reduction in thyroid mass during treatment [19,20] and a few studies have incorporated this effect into the formalism to compute the activity [21–23]. According to these studies the variation in thyroid mass is dependent, in a first estimate, on the activity according to a first-order equation:

$$\frac{dm}{dt} = -k\frac{A(t)}{m} \quad (12)$$

where k is a constant.

By incorporating Equation 6 into Equation 12, the thyroid mass is given by the relation

$$m(t) = \sqrt{\frac{2kA_0\lambda_2}{\alpha_2 - \alpha_1}\left(-\frac{\exp(\alpha_2 t)}{\alpha_2} + \frac{\exp(\alpha_1 t)}{\alpha_1} - \frac{1}{\alpha_1} + \frac{1}{\alpha_2}\right) + m_0^2} \quad (13)$$

where m_0 is the initial mass of the thyroid (i.e., just before radioiodine therapy).

Equation 4 becomes

$$\frac{dD}{dt} = \frac{A(t)Sm_{\text{ph}}}{m(t)}. \quad (14)$$

This is equivalent to

$$\bar{D} = \int_0^{+\infty} \frac{A_0\lambda_2(e^{\alpha_2 t} - e^{\alpha_1 t})Sm_{\text{ph}}}{\sqrt{\frac{2kA_0\lambda_2}{\alpha_2 - \alpha_1}\left(-\frac{e^{\alpha_2 t}}{\alpha_2} + \frac{e^{\alpha_1 t}}{\alpha_1} - \frac{1}{\alpha_1} + \frac{1}{\alpha_2}\right) + m_0^2}} dt. \quad (15)$$

The integration gives

$$\bar{D} = -\frac{Sm_{\text{ph}}}{k}\left[\sqrt{m_0^2 - \frac{2kA_0\lambda_2}{\alpha_1\alpha_2}} - m_0\right]. \quad (16)$$

Finally, the appropriate activity to administer is given by the relation

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$$A_0 = \frac{\alpha_1 \alpha_2}{2k\lambda_2}\left[m_0^2 - \left(m_0 - \frac{k\bar{D}}{Sm_{\mathrm{ph}}}\right)^2\right]. \qquad (17)$$

The constant k is detailed by Traino *et al.* [23] and is given by

$$k = \gamma \frac{m_0^2}{C}$$

where $\gamma = 2.1 \times 10^{-6}$ (established over a cohort of 40 patients [23]), and C is the maximum thyroid uptake.

Experimental data (i.e., the three determinations of the measured thyroid uptake) were fitted with Equation 6 by the least-squares method and were not corrected for radioactive decay because the amount of radioiodine really present in the thyroid was relevant for the dose calculation.

Statistical analysis

Multivariate analysis was performed to determine whether the different methods gave similar therapeutic activity. In order to achieve this, a correlated ANOVA test followed by a Tukey honestly significant difference (HSD) test were used to assess the significance of differences. When needed, a univariate analysis was achieved with the *t*-test. All statistically significant levels were set at 5% ($P < 0.05$).

Results

Thyroid parameters

The mean thyroid mass was 19.4 ± 13.9 g (21.0 ± 14.6 g for women and 11.6 ± 4.2 g for men). The mean uptakes were 36.2 ± 14.6%, 42.8 ± 9.7% and 27.6 ± 6.8%, respectively at 4, 24 and 96 h. Figure 2 shows the fit of the experimental points by means of the two-compartment model.

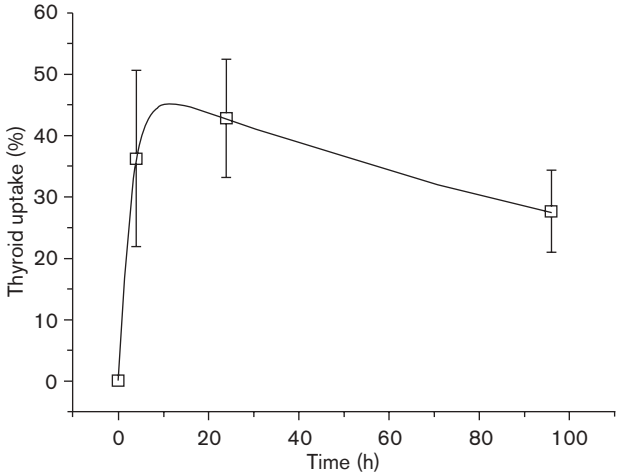
The mean effective half-life was 5.1 ± 1.6 days as calculated by the two-compartment model. The mean effective half-life extracted from the Marinelli 1 method (based on an exponential fit from 24 to 96 h) was 4.8 ± 1.5 days. In this last case, it was impossible to determine the effective half-life for four patients because the computed value exceeded 8 days.

Computation of activity to be administered

Activities were computed with the eight previously described methods for the 41 patients with a prescribed dose of 106 ± 22 Gy depending on the aim of treatment (euthyroidism or hypothyroidism). The mean activities calculated for each method are given in Table 1.

According to the correlated ANOVA test, at least two methods gave statistically different results ($P < 0.001$). The results of the Tukey HSD test are summarized in Table 1. The Leslie 1, Leslie 2 and Bockisch methods gave the lowest computed activities and Marinelli 1 and 2 the highest. Results obtained with Leslie 1 were the lowest and were statistically different from all other methods. Marinelli 1 and 2 differed from, respectively five and six other methods. The Marinelli 3, Leslie 2, Bockisch and approach 2 methods were different from three methods. Approach 1 differed only from two methods and its results were in the same range as the majority of the other methodologies.

Fig. 2



Mean thyroid uptake mean at 4, 24 and 96 h with the corresponding two-compartment modelling. Each experimental point represents the mean thyroid uptake (with the standard deviation bar) over the 41 patients. (□), Experimental data; (–), two-compartment model.

Table 1 Mean activity according to the physician prescribed dose for each formalism

Method	Mean activity (MBq)
Marinelli 1	262 ± 176[a]
Marinelli 2	275 ± 223[b]
Marinelli 3	214 ± 165[c]
Leslie 1	128 ± 95[d]
Leslie 2	193 ± 143[e]
Bockisch	193 ± 137[f]
Approach 1	227 ± 148[g]
Approach 2	213 ± 124[h]

[a] $P < 0.01$ between Marinelli 3, Leslie 1, Leslie 2, Bockisch and approach 2.
[b] $P < 0.01$ between Marinelli 3, Leslie 1, Leslie 2, Bockisch, approach 1 and approach 2.
[c] $P < 0.01$ between Marinelli 1, Marinelli 2 and Leslie 1.
[d] $P < 0.01$ between Marinelli 1, Marinelli 2, Marinelli 3, Leslie 2, Bockisch, approach 1 and approach 2.
[e] $P < 0.01$ between Marinelli 1, Marinelli 2 and Leslie 1.
[f] $P < 0.01$ between Marinelli 1, Marinelli 2 and Leslie 1.
[g] $P < 0.01$ between Marinelli 2 and Leslie 1.
[h] $P < 0.01$ between Marinelli 1, Marinelli 2 and Leslie 1.

Discussion

When comparing the effectiveness of activities calculated by numerous methods it is necessary to include a very large number of patients, who must be assigned to several

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groups according to each calculation method, and matched on numerous parameters including at least age, sex, clinical conditions, thyroid volume and medications. Furthermore, the choice of the amount of radioiodine activity to administer in Graves' disease patients can differ greatly among institutions, depending on the aim of the treatment; that is, to cure hyperthyroidism after one or more treatments or to render the patients either euthyroid or hypothyroid. Therefore, the next point to be considered should be the definition of treatment effectiveness criteria. Eradication of hyperthyroidism could be considered as the only parameter of therapeutic success, but other parameters could be implemented, for instance, related to the risk-to-benefit ratio of radiation with particular concerns about the irradiation of normal tissues. Unfortunately, there is no good indicator of thyroid tissue response to radioiodine therapy, and this latter can vary among individuals, depending on several conflicting factors, including individual thyroid tissue sensitivity, which still remains unknown. As a result, clinical outcome after radioiodine therapy is often disappointing. High rates of hypothyroidism are reported in most series [1,5,6,24,25], whereas over the last decades, much attention has focused on achieving euthyroidism by adjusting the activity of ^{131}I. Thus, although obtaining a sustained euthyroidism appears to be a futile objective, we believe, in line with others, that the optimal approach should be to eradicate hyperthyroidism at the lowest effective radioiodine activity; that is, with an activity calculated by using personalized dosimetry. Moreover, two principal arguments justify a personalized dosimetric approach. Firstly, according to the French regulations laws based on the European Council Directive 97/43 [26], exposures of targets volume shall be individually planned and exposures of non-target volumes shall be as low as reasonably achievable. Secondly, calculating individual biokinetics gives the possibility of providing practical radiation protection guidelines for each patient when he or she leaves the institution. This can be calculated with an appropriate formalism as developed by Cormack and Shearer [27].

Two personalized dosimetric methods for prescribing a therapeutic ^{131}I activity in Graves' disease patients are presented. These two methods (approaches 1 and 2) are based on MIRD methodology. The main assumption was the uniform distribution of ^{131}I in the thyroid gland, which can be considered only for Graves' disease, because of a homogeneous thyroid distribution of radioiodine. The amount of therapeutic activity to administer to a patient is based on a computation with a suitable, weighted, S factor depending on both thyroid mass and patient sex [13], and on three-weekly uptake measurements of ^{131}I (after administration of a low, tracer test activity of 2 MBq before treatment), which allowed an accurate determination of radioiodine biokinetics. The kinetics of ^{131}I in the thyroid and in blood have been evaluated with simplified two-compartment modelling, as suggested by Di Martino *et al.* [17]. Although this modelling was reliable and robust, mathematically speaking, some studies showed variations up to 15% between the tracer test kinetics and the therapeutic kinetics, assumed to be due to an enhanced iodine turnover under therapeutic conditions [11]. However, the same authors reported a strong correlation between the tracer test kinetics and therapeutic kinetics, except for patients whose medication changed close to the test study or therapy.

Finally, although both formalisms take into account the thyroid mass at the time of therapy, the variations of the thyroid volume during the treatment are integrated only into approach 2. In 52 Graves' disease patients, Murakami *et al.* observed a reduction of thyroid volume from 1–6 months after radioiodine therapy related to the outcome of thyroid function [19]. A mathematical model of thyroid mass reduction after ^{131}I therapy was proposed by Traino *et al.*, based on patient thyroid masses evaluated by ultrasonography and applied to a general formula for calculation of the thyroid absorbed dose [22,23]. Thus, reduction of the thyroid mass and absorbed dose to the thyroid can be both predicted and depends on the parameter k established on a cohort of patients. In contrast with the findings of Traino *et al.*, we did not find any significant difference between the results of approach 1 and approach 2; that is, between ^{131}I activity calculated considering a reduction of thyroid mass and that obtained without considering a change in thyroid mass. Such a discrepancy could be explained by the fact that the determination of thyroid mass is one of the intrinsic limitations shared by all methods detailed in this work. This measurement is commonly performed with two-dimensional ultrasonography which can bring a 30% error compared to the real volume of the thyroid [28]. Thus, errors in the volume determination by ultrasonography could lower the difference between approaches 1 and 2, and could at least partially explain the difference between our findings and those obtained by Traino *et al.* However, our methods reflect clinical practice, in which two-dimensional ultrasonography is most often performed without any quality control or calibration of the ultrasound machine. Other methods may be more reliable or accurate, such as three-dimensional ultrasonography or single photon emission computed tomography, but are still not implemented in routine [28,29].

Our results showed that the two personalized dosimetric methods (approaches 1 and 2) were not statistically different from Marinelli 3 (using only a measurement at 24 h), Leslie 2 (using only a measurement at 24 h) and Bockisch (using only a measurement at 96 h). It could be argued that personalized dosimetry with one uptake

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Table 2 Ratio deviation for the 41 patients (minimum and maximum values) between activity calculated by the so-called Marinelli 1, Marinelli 3 and Bockisch and activity computed by approach 1

Ratio	Minimum–maximum
Marinelli 1/approach 1	10.8–1.9
Marinelli 3/approach 1	0.5–1.4
Leslie 2/approach 1	0.3–1.5
Bockisch/approach 1	0.6–1.1

measurement point (24 h or 96 h) was equivalent to a methodology using three uptake measurement points, which is more constraining for the patient. However, this absence of difference can be partly explained by comparing effective half-lives and maximum uptakes. Indeed, the average effective half-life derived from experimental uptake measurements over the 41 patients was 5.1 ± 1.6 days using two-compartment modelling. This value was close to the fixed value of 5 days used in the Marinelli 3 methodology. On the other hand, the mean of the maximum uptake (as determined by the two-compartment model) was similar to the mean of the maximum uptake at 24 h (successively used in the Marinelli 3, Leslie 1 and Leslie 2 methods). The mathematical fit of experimental data gave a value of 0.45 ± 0.12 which was similar to that given at 24 h, i.e. 0.43 ± 0.09 (a Student's *t*-test did not show any significant difference). However, the absence of a statistical difference between our formalism and the other methods must be moderated by the presence of large intra-individual differences. For a given patient, it was possible to obtain a ratio of up to 2 (respectively, 0.5) between two methods statistically equivalent (Table 2). These findings were in agreement with previously published data with an average ratio between the two methods which could reach 2.5 [30].

Conclusion

Our approaches are more constraining for the patient because three time points (with one at 96 h) need to be measured. However, these new MIRD-based formalisms appear to be a good compromise between the common formalisms already used in many institutions, with the main advantage that they take into account the individual biokinetics and the reduction of thyroid mass after therapy (only for the approach 2). Furthermore, these personalized dosimetries allow the exposures of target volumes and non-target volumes to be planned individually, and practical radiation protection guidelines to be adapted for each patient when he or she leaves the institution. Nevertheless, further studies are required to establish the effectiveness of these new methods, in particular their ability to eradicate hyperthyroidism after only one personalized therapeutic ^{131}I activity (i.e., without later relapses).

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